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Summary

- Lead quant strategist, 13+ years experience, managing quant teams to oversee \$100+bn multi-asset portfolios
- Statistics, machine learning & AI for portfolio analytics, risk models and global markets quant research

Professional Experience

GIC Lead Quant Strategist, Senior Vice President, Global Markets Multi-Assets Singapore, since 2023

- Lead and mentor a team of six quants on portfolio management and risk models, quant market research
- Design and implement portfolio construction and risk models for \$100+bn multi-asset portfolios
- Architect portfolio risk tools: CVaR & DD limits, risk factors, stress-tests, early warning signals
- Steer quant research using Machine Learning/AI/LLM on rates, FX, and credit markets
- Design cloud platforms, automatize performance & risk reporting and engage with decision makers

ADIA Senior Quant Researcher, Portfolio Analytics & Market Research Abu Dhabi, 2022-2023

- Portfolio construction and risk models, supporting CIO tactical tilts and FX / duration hedging
- Advanced macro research across global rates, FX, and commodity to identify opportunities and risk

International Monetary Fund Quant Financial Sector Expert Washington DC, 2016-2022

- Bank stress-tests (FSAP) and technical assistance missions on quant modeling in 30+ EM countries
 - Solvency and liquidity stress-tests, capital adequacy, macro models for risk scenarios (Growth-at-Risk)
 - Risk and liquidity management frameworks, FX reserves management tools, financial stress indices
- Analytics: sovereign risk, macro stress-tests, nowcasting, inflation forecasting, debt projections

European Central Bank Portfolio Manager and Quant Researcher Frankfurt, 2013-2016

- Managed RMB and JPY reserve portfolios and supported QE market operations
 - Portfolio management and risk modeling for ECB rates and FX portfolios

Education

Ph.D. Quantitative Finance, Peking University Beijing 2013

Entire curriculum completed in Chinese Mandarin in Beijing

M.Sc. Mathematics and Statistics, ENSAE Paris Paris 2007

Skills

- **Analytics:** Statistics & machine learning, AI & LLM, convex optimization, bootstrapping, density models
- **Tech stack:** Python, Github CI/CD, cloud computing (AWS & Databricks), Airflow schedulers, big data
- **Data:** Bloomberg, Macrobond, Haver, CEIC, Wind, Factset, alternative datasets
- **Risk models:** VaR/CVaR, stress-test, early-warning signals, rates & FX risk, sovereign risk, FSI
- **Risk factors systems:** Blackrock BRS, MSCI Barra, Axioma, PCA-custom risk factors
- **Languages:** English (fluent), Chinese (conversational), French (native)

Python Packages

- Distributional GARCH model to design VaR-based FX Interventions for Central Banks, [varfxi repo](#)
- Density Forecasting and Risk Modeling for Macro Series, [gar repo](#)
- Conditional Density Projection via Quantile Regressions, Resampling and Multifit Models, [quantproj repo](#)
- Robust Density Estimation via Over-Parametrized Gaussian Skewed Models, [robustdensity repo](#)
- Partial Least Squares Wrapper for Data Reduction based on Scikit, [plswrapper repo](#)
- Quantile Spacing- solving for quantile crossing, from Schmidt and Zhu (2016), [quantilespacing repo](#)
- Quantile Local Projections, [quantileproj repo](#)
- Granular Instrumental Variables - from Gabaix and Koijen (2020), [granulariv repo](#)
- Cluster Analysis Wrapper with Performance Metrics and Visualization Tools, [clusterwrapper repo](#)

Academic Research

- *Financial Conditions in Europe: Dynamics, Drivers, and Macroeconomic Implications*, [IMF WP \(2024\)](#)
- *Technology and the Geography of FX Markets*, [Journal of International Money and Finance \(2023\)](#)
- *FX Interventions: A Risk-Based Framework*, [IMF WP \(2020\)](#)
- *Predictive Density for Global Growth*, [IMF WP \(2020\)](#)
- *Growth at Risk: Applications in IMF Surveillance*, [IMF WP \(2019\)](#)
- *Can Countries Manage Their Financial Conditions Amid Globalization?*, [IMF WP \(2018\)](#)
- *Trading Without a View: Fast Trading, Entropy and Underreaction in the FX Market*, [ECB WP \(2018\)](#)
- *Household Savings in China : The Role of the Housing Market and Borrowing Constraints*, [BdF WP \(2014\)](#)

Policy Research

- *United Arab Emirates: TA Report - Liquidity Management and Forecasting*: [IMF TA report \(2022\)](#)
- *Jordan: TA Report - Forecasting Framework for Currency in Circulation*: [IMF TA Report \(2022\)](#)
- *House Price Synchronization: What Role for Financial Factors?*: [IMF GFSR \(2018\)](#)
- *Financial Conditions and Growth at Risk*: [IMF GFSR \(2017\)](#)
- *High-Frequency Trading, Information and Market Volatility: the Role of Dark Pools*: [ECB FSR \(2017\)](#)
- *The Role of Currency Invoicing for the Transmission of Exchange Rate Movements*: [ECB IRE \(2015\)](#)

Country Missions

Policy experience with 30+ country missions on technical assistance, policy negotiations and financial surveillance

Asia, Europe and LATAM

- **P.R. China:** Financial stress models (Mission Chief, 2019)
- **P.R. China:** ECB negotiations, RMB Eurosystem portfolio (2015)
- **India:** Sovereign risk models (2019)
- **South Korea:** FX reserves management (2020)
- **Hong Kong SAR:** FSAP, systemic liquidity and stress-testing (2021)
- **Philippines:** Sovereign risk models (2019)
- **Malaysia:** Sovereign risk models (2019)
- **Singapore:** Sovereign risk models (2019)
- **Albania:** IMF Article IV and post-program (2018, 2019)
- **Bosnia:** FX management (Mission Chief, 2022)
- **Peru:** FSAP, systemic liquidity and stress-testing (2017, 2018)

MENA and SSA

- **United Arab Emirates:** central bank operations (co-Mission Chief, 2021)
- **Jordan:** liquidity forecasting (Mission Chief, 2021)
- **Israel:** IMF Article IV, external sector (2019)
- **Morocco:** FX interventions (2020)
- **Algeria:** central bank operations (Mission Chief, 2022)
- **Tunisia:** liquidity forecasting (2020)
- **WAEMU:** FSAP, stress-testing (2021)
- **Mauritania:** central bank operations (2023)
- **Namibia:** liquidity forecasting (Mission Chief, 2021)
- **Djibouti:** liquidity forecasting (Mission Chief, 2021)
- **D.R Congo:** IMF program negotiations (2020)
- **Burundi:** central bank operations (2022)

Teaching

I taught financial modeling and financial operations to countries officials and central bankers around the world, often via IMF technical assistance centers.

- **Singapore:** [FX Intervention Rules](#) (2023)
- **Singapore:** liquidity forecasting (2022)
- **Ghana:** liquidity forecasting (2022)
- **Kuwait:** liquidity forecasting (2021, 2022)
- **Slovenia:** liquidity forecasting (2022)
- **IMF Washington DC:** density modeling (2021)
- **Kazakhstan:** liquidity forecasting (2021)
- **Guatemala:** liquidity forecasting (2021)
- **Austria:** density modeling (2021)
- **Malaysia:** density modeling (2020)
- **IMF Washington DC:** Python (2019)
- **Sciences Po and ENSAE Paris:** Econometrics and macroeconomics (2010-2012)